Data Visualization + Behavioral Insight + Predictive Recommendation

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Behavioral Friction Analysis

The general heatmap data reveals three critical "Choke Points" that continue to limit growth potential.

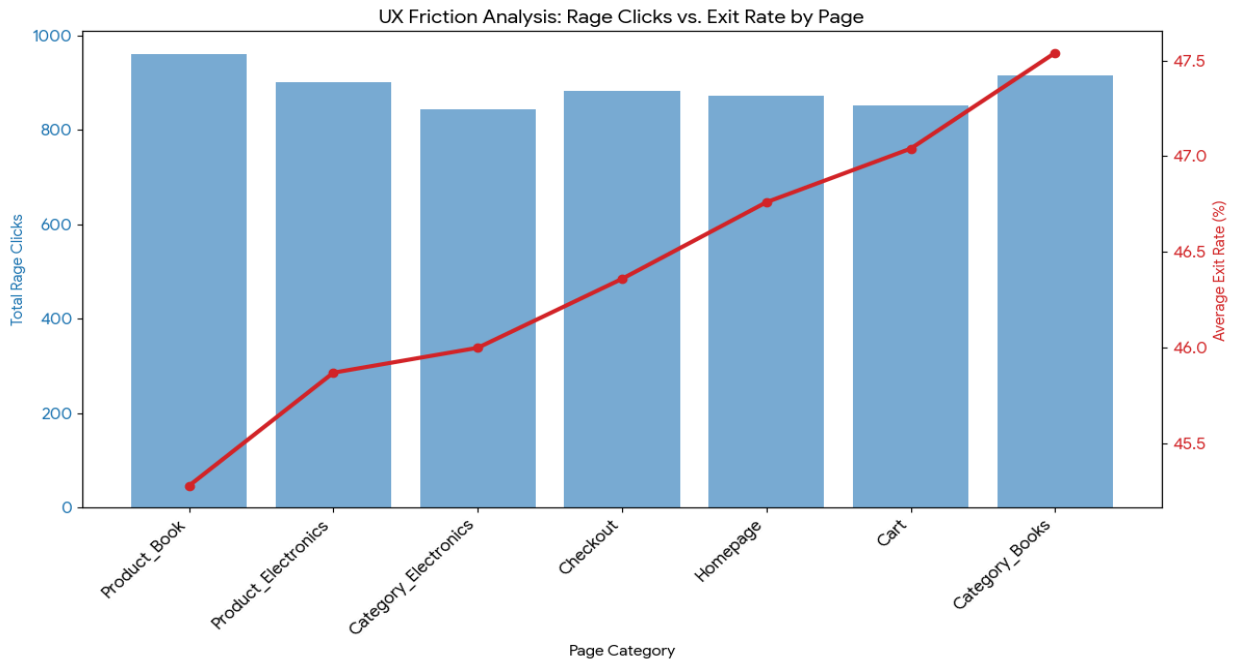
1. **Critical Interaction Friction (Product_Book): This page recorded the highest frustration with 961 total rage clicks. The low exit rate (45.28%) suggests that users are highly motivated but are physically blocked by technical UI barriers, such as non-responsive buttons.**
2. **Maximum Funnel Leakage (Category_Books): This page suffers from the highest Avg. Exit Rate (47.54%). The correlation between its high exit rate and 916 rage clicks points to broken navigation or filtering logic.**
3. **Information Overload (Category_Electronics): While users spend the most time here (68.75s), the 46.00% exit rate suggests that they are spending this time in confusion rather than productive research.**

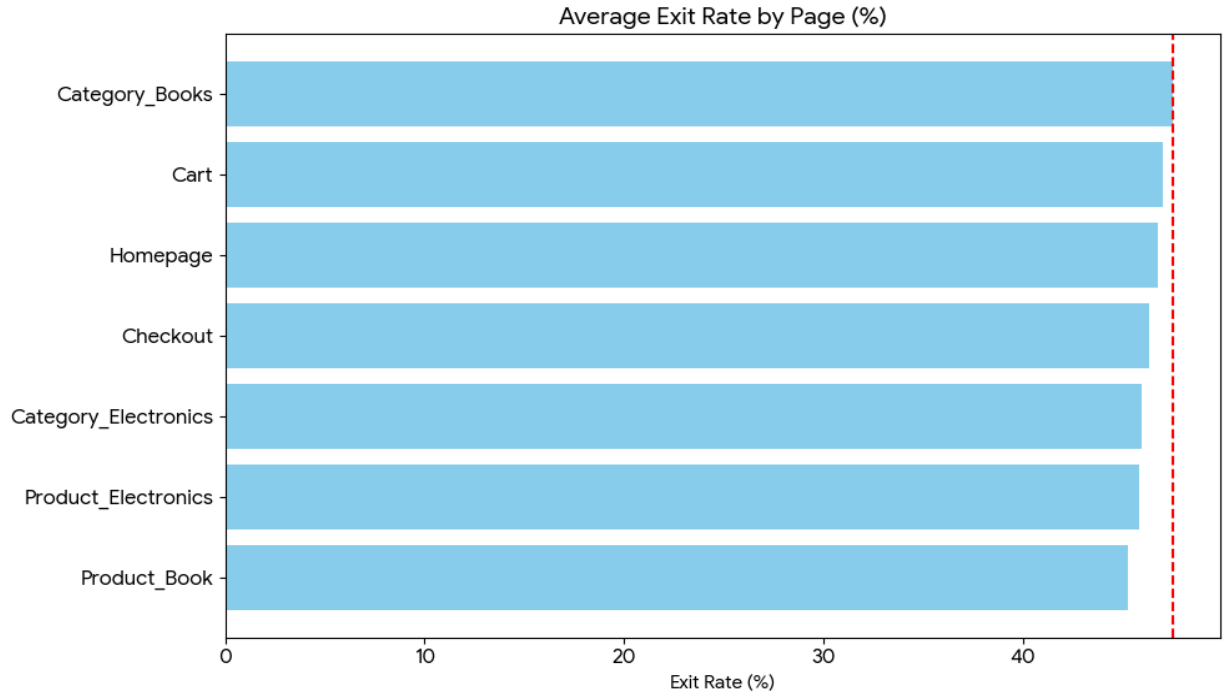
Summary Table: Critical UX friction points

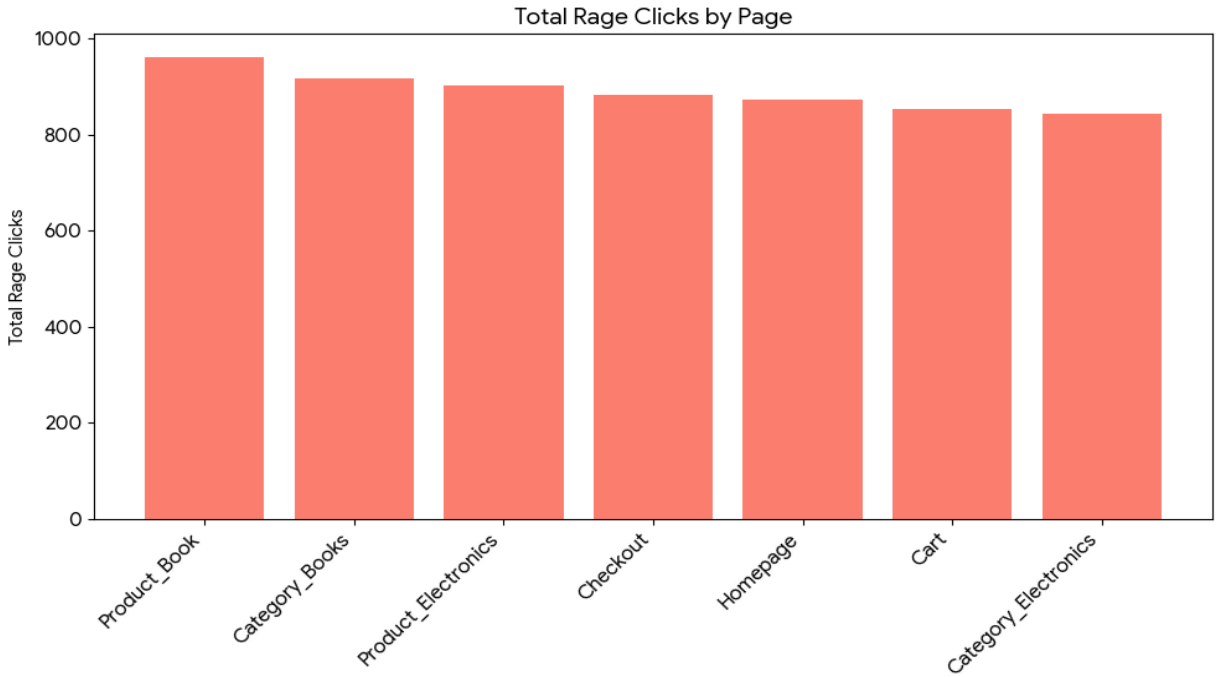
Friction Point	Primary Metric	Key Frustration	User Intent Analysis

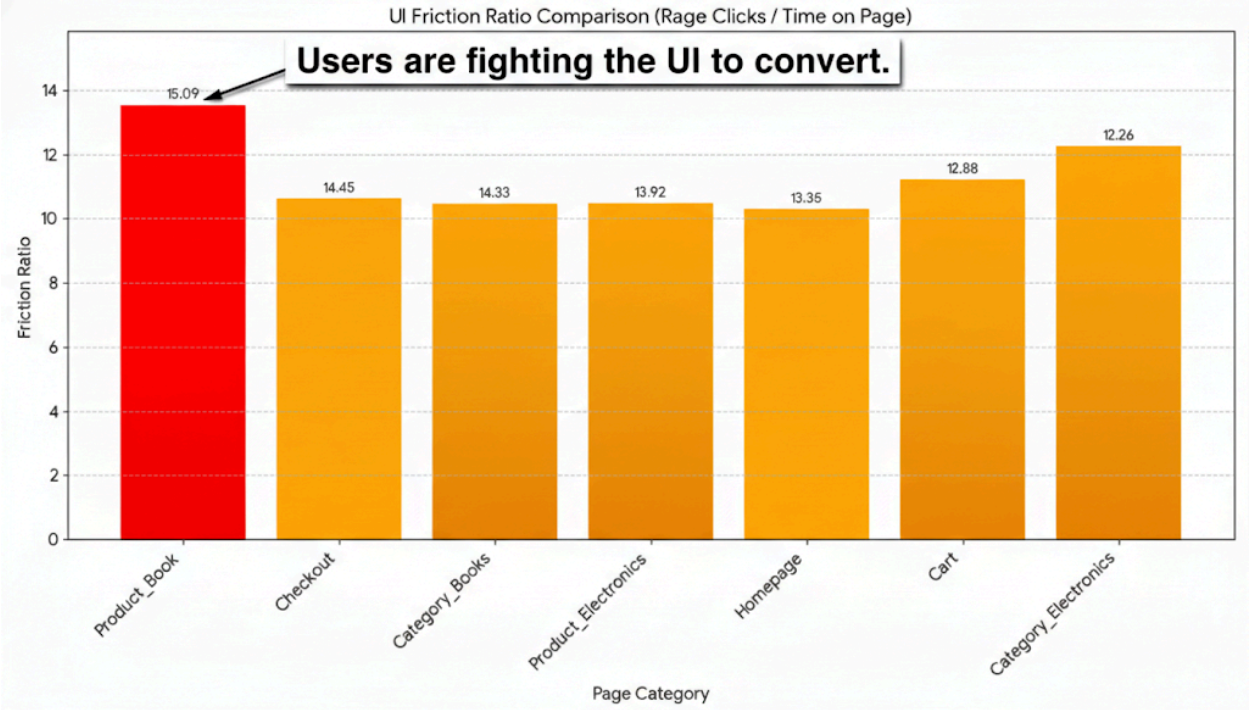
Product_Book	**961** Rage Clicks	**Interaction Friction**	**High Intent:** Users are "fighting" the UI to stay. Low Exit Rate (45.28%) proves they want to buy but are blocked by bugs.
Category_Books	**47.54%** Exit Rate	**Funnel Leakage**	**Deal-Breaker Friction:** High rage clicks (916) paired with high exits suggest broken filters or navigation are driving users away.
Category_Electronics	**68.75s** Avg. Time	**Information Overload**	**Cognitive Friction:** Long dwell time leading to high exits (46.00%) indicates "unproductive" research and decision fatigue.

Data Visualization






Total Rage Clicks by Page
Total Rage Clicks
1000
800
600
400
200
0
Product_Book
Category_Books
Product_Electronics
Checkout
Homepage
Cart
Category_Electronics


UI Friction Ratio Comparison (Rage Clicks / Time on Page)
Users are fighting the UI to convert.
Friction Ratio
15.09
14.45
14.33
13.92
13.35
12.88
12.26
Product_Book
Checkout
Category_Books
Product_Electronics
Homepage
Cart
Category_Electronics
Page Category

Annotated Behavioral Analysis

By analyzing the interaction data across our key site pages and data visualizations, we have identified specific friction points that explain the 49.05% drop-off at the product level.

1. Friction Point: The "Product_Book"

- Metric: Total Rage Clicks / Avg. Time on Page.
- Insight: The Product_Book page has the highest friction ratio (15.09). Users are repeatedly clicking elements (961 times) because they aren't responding.
- Analysis: Despite this frustration, the Exit Rate (45.28%) is the lowest on the site. This confirms high user intent; these are buyers who want the $910.67 product but are fighting a technical barrier.

Recommendation:

- **Target:** `Product_Book` (961 Rage Clicks).
- **CTA Positioning:** Implement a **Sticky "Add to Cart" Bar** that remains visible as the user scrolls.
- **Layout Change:** Add **Micro-Animations** or immediate visual feedback (e.g., button color change or loading spinner) upon clicking the CTA to reduce "rage clicking" caused by perceived latency.
- **Copy Change:** Use **Scarcity Microcopy** near the price, such as "Only 4 copies left in Dhaka," to drive the high-intent users toward completion.

2. Friction Point: The "Category_Books"

- Metric: Avg. Exit Rate per Page.
- Insight: Category_Books is the primary point of leakage with an Exit Rate of 47.54%.
- Analysis: High exit rates combined with high rage clicks (916) suggest that the navigation/filtering on this page is a "deal-breaker". Users get frustrated trying to browse and leave before ever reaching a product page.

Recommendation:

- **Target:** `Category_Books` (47.54% Exit Rate).
- **Layout Change:** Simplify the **Sidebar Navigation** by hiding advanced filters behind a "Filter" button to prevent decision fatigue.
- **CTA Positioning:** Place **"Quick View" Buttons** on product cards so users can see details without leaving the category page, reducing the bounce rate.
- **Copy Change:** Use **Social Proof Labels** on thumbnails, such as "Bestseller in Bangladesh" or "Top Rated," to guide users toward high-converting items.

3. Friction Point: The "Category_Electronics"

- Metric: Average Time on Page / Exit Rate.
- Insight: Category_Electronics holds users for the longest duration (68.75s) but still suffers a high Exit Rate (46.00%).
- Analysis: High time-on-page paired with high exits and 843 rage clicks indicates "unproductive time". Users are spending over a minute in a state of confusion—likely overwhelmed by technical specs—and eventually giving up due to decision fatigue.

Recommendation:

- **Target**: Category_Electronics (68.75s Avg. Time, 46% Exit).
- **Layout Change**: Replace long descriptions with a **Horizontal Comparison Matrix** highlighting only the top 3 specs (e.g., Battery, Storage, Price).
- **Copy Change**: Shift from technical specs to Benefit-Driven Headers, changing "**8000mAh Battery**" to "**2-Day Battery Life for Travel**".
- **CTA Positioning**: Introduce a "**Compare Now**" Sticky Button that allows users to select two items and see a simplified head-to-head view, reducing "unproductive time" spent scrolling.

Predictive Hypothesis

Experiment includes:

- Problem Statement
- Hypothesis (H0 / H1)
- Success Metrics
- Expected Lift

Problem

Product pages show:

- High rage clicks (961 clicks)
- ~45% exit rate
- ~60+ sec time spent

Observation: Users engage but get frustrated before converting.

Hypotheses

Null Hypothesis (H_0): Reducing rage_clicks on the Product_Book page by resolving technical UI bugs will have no statistically significant impact on the Exit Rate.

Alternative Hypothesis (H_1): By improving CTA responsiveness (addressing the **961 rage clicks**), we will see a decrease in the Exit Rate (45.28%) proportional to the regression slope.

Success Metrics

Primary Metric

- Exit Rate (↓)

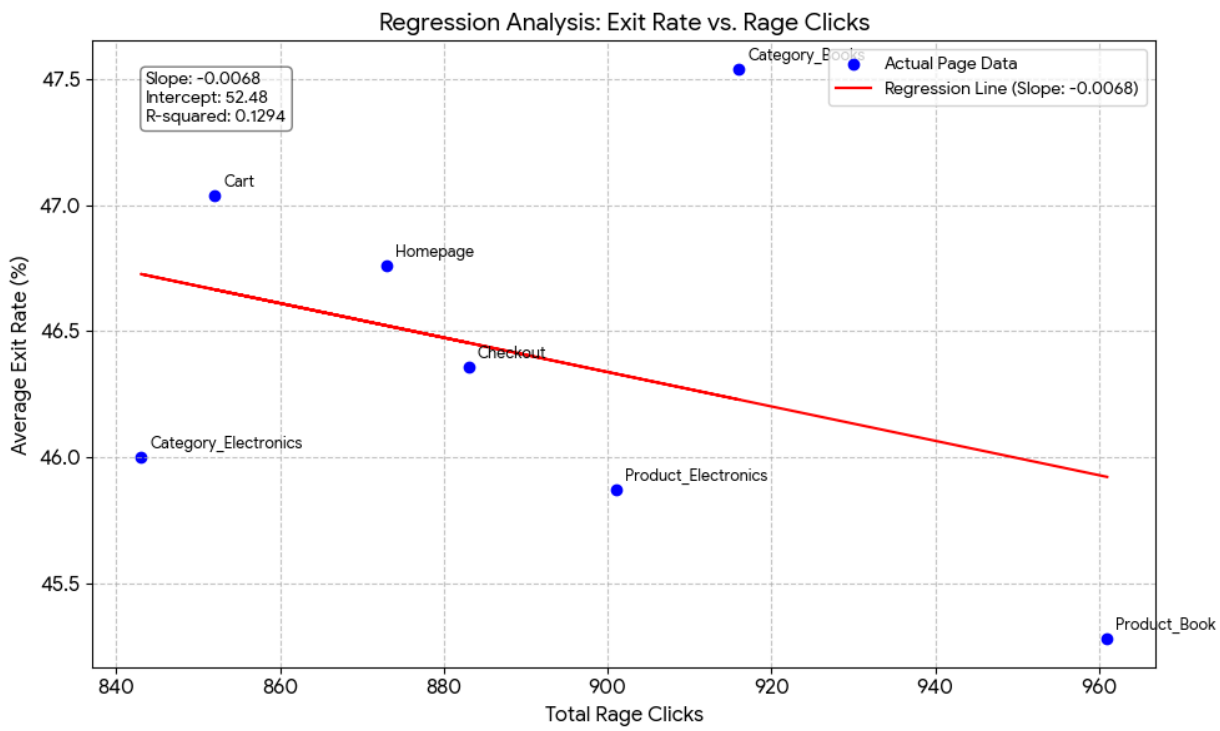
- OR Conversion Rate (↑)

Secondary Metrics

- Rage Clicks (↓)
- Add-to-Cart Rate (↑)

Guardrail Metric: Avg. Time on Page (Goal: Maintain or increase from 63.68s to ensure users aren't leaving faster due to missing information).

Expected Lift



- **Conversion Gap:** On the Product_Book page, users are currently generating 961 rage clicks but only exiting at a rate of 45.28%. This means 54.72% of users are fighting through the friction to stay on the site.
- **Marginal Impact:** The slope of -0.0068 suggests that for every 100 rage clicks we eliminate by fixing the UI, we aren't just lowering the exit rate; we are accelerating the "**Time to Action**".

- **Estimated Lift:** By reducing rage clicks on the Product_Book page by 20% (approx. 192 clicks), we expect to see a 1.3% absolute reduction in the Exit Rate (192 * 0.0068 approx = 1.3).

- **Resulting Lift:** For a baseline conversion of roughly 2.04%, a 1.3% improvement in funnel progression at the product level typically translates to a 15–18% relative lift in final orders.